

PriceRight

An accountability layer for x402 agent commerce on Base

signed x402 payments · ERC-8004 identity + collateral · deterministic slashing

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The problem

x402 makes an agent-to-agent USDC payment on Base one HTTP round trip.

But the rail prices the **request**, not the **answer**:

- answer right, keep the fee
- answer **wrong**, keep the fee anyway

The lazy resolver reads a stale cache instead of the fresh check, collects, and is wrong. The party who built a workflow on that answer pays for it.

x402 priced the request. **Nothing prices being wrong.**

Why Base, why now

Three things are already true on Base at once:

- **x402** settlement is native: the `exact` scheme is USDC's EIP-3009 `transferWithAuthorization`, a fee bound to a unit of work
- **CDP AgentKit** gives agents wallets that can hold collateral and sign
- **ERC-8004** identity carries reputation that travels when the identity is sold

The accountability layer is the missing brick. PriceRight is it.

Scope (milestone 1)

One economy, one hero, fully built and tested:

- a resolver agent answers subjective **yes/no** tasks
- it **pays** to answer with x402 and **bonds its ERC-8004 identity**
- a **deterministic** settlement rewards right and **slashes** wrong

Base Sepolia, testnet only. Not a marketplace, not a DAO. One credible
punishment,
end to end.

The solution

A task poster commits a hidden ground truth.

1. the claim endpoint answers **402 Payment Required**
2. the resolver signs an **EIP-3009 authorisation** and sends **X-PAYMENT**
3. collateral is bonded against its **ERC-8004 identity**
4. it commits a verdict + **keccak-256 of its reasoning**
5. settlement compares committed verdict vs revealed truth

right → fee back + bounty + collateral, feedback 100

wrong → collateral seized to the poster, feedback 0

How it works

```
postTask(bounty, fee, collateral, keccak(truth,salt))      poster
claimTask(taskId, agentId, X402Payment{...v,r,s})         resolver
  -> token.transferWithAuthorization(...)   fee moves by signature
  -> vault.bondFor(bondKey(taskId), agentId, ...)
commitVerdict(taskId, v, keccak(reasoning))               resolver
settle(taskId, truth, salt)                               poster
  correct -> vault.release + credit resolver + feedback 100
  wrong   -> vault.slash + credit poster + feedback 0
```

The slash is a **pure function** of (verdict, truth). No override, and no path leaves collateral stranded in the contracts.

Demo

Keyless, no node, no credentials:

```
PRICERIGHT_OFFLINE=1 python -m priceright.main demo
```

- **agent #1** evidence policy → **CORRECT, +110 tUSD, collateral returned, score 100%**
- **agent #2** stale-cache policy → **WRONG, 50 tUSD collateral seized, score 0%**

Point `PRICERIGHT_RPC_URL` at Base Sepolia and the same code path sends real transactions. Agent #2 is not rigged to lose: give its cache a threshold it clears and it wins.

Why the sponsor tech is load-bearing

- **x402**: the real `exact` scheme - 402 challenge, EIP-3009 authorisation over USDC, verify, settle. The fee moves by signature with **no allowance** to the arena.
- **CDP AgentKit**: the resolver and poster identities are CDP wallets (M2); the signer sits behind a seam so the swap does not touch the arena.
- **ERC-8004**: identity is a **real ERC-721** (`supportsInterface` 0x80ac58cd); feedback is permissionless and client-filtered; reputation follows the token.
- **Base**: Base Sepolia deployment and a queryable public slash are the M1 artifact.

Results

- **46** Foundry tests and **53** Python tests pass, from a clean checkout
- The Python x402 signature is verified by Solidity `ecrecover`, and the Python suite regenerates that vector and pins it against the `.sol` constants - neither side drifts
- The remote facilitator is a real HTTP round trip that **fails closed**: refused or unreachable leaves the task open, nothing paid
- Every terminal path empties the escrow; the timeout paths are permissionless

Not yet on Base Sepolia (no funded key here); `docs/HONESTY.md` and `docs/LIMITATIONS.md` say so plainly.

Roadmap

- **M1** PriceRight repo + Base Sepolia deployment + a recorded public slash
- **M2** CDP AgentKit wallets as the resolver and poster identities
- **M3** settlement through the hosted CDP facilitator, same fail-closed guarantee
- **M4** a `bond / claim / commit / settle / readReputation` SDK, reference agent, docs

Open source, MIT, testnet-scoped. A mainnet audit precedes any real-value use.

PriceRight

Pay to answer. Bond your identity. Be wrong, pay for it.

right is paid · wrong is seized · no human in the loop

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